

PROBLEM SET 5: MATCHING MODELS

PROBLEM I – A MATCHING MODEL

We consider the following matching model. The matching function is given by

$$m(u, v) = \frac{uv}{u + v}. \quad (1)$$

There is free entry in vacancy posting and the cost of posting a vacancy per unit of time is equal to c . Wages are set by intertemporal Nash bargaining so that

$$V_e = V_u + \varphi W, \quad (2)$$

and

$$J = (1 - \varphi)W \quad (3)$$

The job destruction rate is s and the real interest rate is r .

1 – What is the function $q(\theta)$? Check that $m(u, v)$ has constant returns to scale and is concave in each of its arguments. What is the elasticity of m with respect to u as a function of θ ?

2 – Show that the Beveridge curve has the following equation:

$$\theta = \frac{s(1 - u)}{u - s(1 - u)} \quad (4)$$

3 – Show that the equilibrium value of θ must satisfy:

$$y/c = \frac{r + s}{1 - \varphi}(1 + \theta) + \frac{\varphi}{1 - \varphi}\theta \quad (5)$$

4 – What is the steady-state unemployment rate u ? How does it depend on r, s, c, y , and φ ? Why?

5 – Show that the following relationship exists between the wage w and θ :

$$w = \frac{\varphi c}{1 - \varphi}[(r + s)(1 + \theta) + \theta] \quad (6)$$

6 – How do wages depend on r, s, c, φ, θ according to this formula? Explain?

7 – What would be the observed correlation between wages and unemployment if the driving force for changes in unemployment was:

- changes in productivity ?
- changes in the discount rate?
- changes in the cost of posting vacancies?
- changes in workers' bargaining power?

8 – Write down the central planner's problem, the Hamiltonian, and the corresponding first-order conditions.

9 – Show that along the optimal path the marginal value of an extra job λ is related to θ according to:

$$\lambda = c(1 + \theta)^2. \quad (7)$$

Explain the economic intuition for why it increases with c and θ .

10 – Show that in steady state the optimal value of θ must satisfy

$$y/c = (r + s)(1 + \theta)^2 + c\theta^2 \quad (8)$$

11 – What relationship must hold among the model's parameters for (8) to have the same solution as (5) ?

12 – Check that this is equivalent to

$$\varphi = \frac{\theta}{1 + \theta}, \quad (9)$$

with θ being the common solution to these two equations.

13 – How does this last condition relate to the Hosios efficiency condition?

PROBLEM II – ANOTHER MATCHING MODEL

Consider the following version of the matching function giving the number of matches M :

$$M = aV^\alpha U^{1-\alpha} \quad (1)$$

where V is the number of vacancies and U is the number of unemployed job-seekers. Assume that $\alpha = 1/2$.

1 – Define $\theta = \frac{V}{U}$ as the measure of labor market tightness. By using the matching function (1), write the rate at which a firm fills a vacant job $\frac{M(V,U)}{V}$ and the rate at which an unemployed worker finds a job $\frac{M(V,U)}{U}$ as a function of θ . Denote the former with $q(\theta)$ and the latter with $p(\theta)$.

2 – Compute $q'(\theta)$ and $p'(\theta)$. Interpret economically the signs of these functions.

3 – Denote the exogenous job destruction rate with δ , wage with w , non-labor market income with b , and the discount rate with r . Write the discounted expected utility of an employed person V_e and the discounted expected utility of an unemployed person V_u where you should use the exit rate from unemployment derived previously.

4 – Denote the output by worker with y and the cost of holding a vacancy with c . Write the discounted expected profits from a filled job Π_e and the discounted expected profits from a vacant job Π_v where you should use the probability of a match occurring derived previously.

5 – Assuming free entry, use the expressions for Π_e and Π_v to derive the firm's labor demand that gives the relationship between θ and the net production $y - w$. This gives the labor demand equation (LD).

6 – Solve the Nash bargaining problem:

$$\max_w \Omega = \frac{1}{2} \log(V_e - V_u) + \frac{1}{2} \log(\Pi_e - \Pi_v)$$

assuming free entry. Write the wage as a function of θ . This gives the wage equation (WE).

7 – Solve for the steady state unemployment where flows in and flows out from unemployment are equal. Use the formulas for $p(\theta)$ and $q(\theta)$ derived previously. This gives the Beveridge curve equation (BC).

8 – By combining the equations (LD) and (WE), one obtains an equation that relates θ to y , b , c , r , δ and a . Derive this equation.

9 – Show how θ , U and w vary with b . Interpret the answers.

10 – Show how θ , U and w vary with a . Interpret the answers.